

Computational Companion to “Generalized Quasi-Symmetric Nets: A Constructive Approach to the Equimodular Elliptic Type of $n \times m$ Kokotsakis Polyhedra” — Helper

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Tested on: Mathematica 14.0

Section 1

Results of Section 1 are used in the proof of Lemma 8 (see paper)

In[113]:=

```
(* =====*)
(*PART 1:INITIALIZATION& DOMAIN ASSUMPTIONS*)
(* =====*)
ClearAll[ $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$ ,  $\sigma$ ,  $\varepsilon$ , a, b, c, d, r, s, f, M, rad,
  sq, assumptions, isTrueQ, isTrueQeps, row, identities];
 $\sigma = (\alpha + \beta + \gamma + \delta) / 2$ ;
a = Sin[ $\alpha$ ] / Sin[ $\sigma - \alpha$ ]; b = Sin[ $\beta$ ] / Sin[ $\sigma - \beta$ ];
c = Sin[ $\gamma$ ] / Sin[ $\sigma - \gamma$ ]; d = Sin[ $\delta$ ] / Sin[ $\sigma - \delta$ ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain
  assumptions:the flat angles and the sigma-shifted angles lie in (0,Pi);
  these hold under the hypotheses of the lemma
  indicated in the subsection above (see paper)*)
assumptions = {0 <  $\alpha$  < Pi, 0 <  $\beta$  < Pi, 0 <  $\gamma$  < Pi,
  0 <  $\delta$  < Pi, 0 <  $\sigma - \alpha$  < Pi, 0 <  $\sigma - \beta$  < Pi, 0 <  $\sigma - \gamma$  < Pi, 0 <  $\sigma - \delta$  < Pi};
rad = (f - M) (r - 1) (s - 1) / (f (1 - M));
sq = (Sin[ $\sigma$ ] / Sin[ $\sigma - \delta$ ]) ^2;
```

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isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions]];
(*Two-stage verification,
in the order of the table:STAGE 1 (the radicand row) identifies rad with the
  perfect square sq.STAGE 2 (the identity row) uses Stage 1:for t=
  Sin[sigma]/Sin[sigma-delta],the eps-split assumptions
  give t the sign eps (u/v checks),
so Sqrt[rad]=Sqrt[t^2]=eps*t (abstract root law),
and eps*Sqrt[rad]*Sin[sigma-delta]=
  eps^2*Sin[sigma]=Sin[sigma] (cancellation).No Sqrt
  over trigonometric arguments reaches the kernel.*)
isTrueQeps[lhs_, rhs_] :=
  Module[{t, u, v}, TrueQ[FullSimplify[rad == sq, Assumptions → assumptions]] &&
    TrueQ[Assuming[u > 0 && v > 0, Simplify[u / v > 0]]] &&
    TrueQ[Assuming[u < 0 && v > 0, Simplify[u / v < 0]]] &&
    TrueQ[Assuming[t > 0, Simplify[Sqrt[t^2] == t]]] &&
    TrueQ[Assuming[t < 0, Simplify[Sqrt[t^2] == -t]]] &&
    TrueQ[Simplify[(Sin[sigma] / Sin[sigma - delta]) Sin[sigma - delta] == Sin[sigma]]];
(* =====*)
(*PART 2:THE RADICAND AND THE IDENTITY*)
(* =====*)
identities = {"Radicand equals a perfect square",
  HoldForm[(f - M) (r - 1) (s - 1) / (f (1 - M))], HoldForm[(Sin[sigma] / Sin[sigma - delta]) ^ 2],
  isTrueQ}, {"Identity for sin(sigma) (via the radicand row)", HoldForm[Sin[sigma]],
  HoldForm[epsilon Sqrt[(f - M) (r - 1) (s - 1) / (f (1 - M))] Sin[sigma - delta], isTrueQeps}};
(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row[{desc_, lhs_, rhs_, test_}] :=
  {desc, Item[TraditionalForm[lhs], Alignment → Center],
  Item[TraditionalForm[rhs], Alignment → Center],
  If[test[ReleaseHold@lhs, ReleaseHold@rhs],
    Style["Verified", Darker@Green, Bold], Style["FAILED", Red, Bold]]};
Grid[Join[{{Style["Description", Bold], Style["Left Hand Side (LHS)", Bold],
  Style["Right Hand Side (RHS)", Bold], Style["Equality Status", Bold]}},
  row/@identities], Frame → All, Background → {None, {1 → LightGray}},
  Spacings → {2, 2.5}, Alignment → {{Left, Center, Center, Center}},
  ItemStyle → Directive[FontFamily → "Helvetica", 12]]

```

Out[125]=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Radicand equals a perfect square	$\frac{(f-M)(r-1)(s-1)}{f(1-M)}$	$\left(\frac{\sin(\sigma)}{\sin(\sigma-\delta)}\right)^2$	Verified
Identity for $\sin(\sigma)$ (via the radicand row)	$\sin(\sigma)$	$\varepsilon \sqrt{\frac{(f-M)(r-1)(s-1)}{f(1-M)}} \sin(\sigma-\delta)$	Verified

Section 2

Results of Section 2 are used in the proof of Lemma 12 (see paper)

In[146]:=

```
(* =====*)
(*PART 1:INITIALIZATION& DOMAIN ASSUMPTIONS*)
(* =====*)ClearAll[α,
  β, γ, δ, σ, a, b, c, d, r, s, f, M, assumptions, isTrueQ, row, identities];
σ = (α + β + γ + δ) / 2;
a = Sin[α] / Sin[σ - α]; b = Sin[β] / Sin[σ - β];
c = Sin[γ] / Sin[σ - γ]; d = Sin[δ] / Sin[σ - δ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain assumptions:the flat angles lie in (0,Pi)*)
assumptions = {0 < α < Pi, 0 < β < Pi, 0 < γ < Pi, 0 < δ < Pi};
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions]];
(* =====*)
(*PART 2:THE SIX RATIO IDENTITIES*)
(* =====*)
identities = {"Identity 1", HoldForm[Sin[α] Sin[δ] / (Sin[β] Sin[γ])],
  HoldForm[r (r - M) / (M (r - 1))], isTrueQ},
{"Identity 2", HoldForm[Sin[σ - α] Sin[σ - δ] / (Sin[σ - β] Sin[σ - γ])],
  HoldForm[(r - M) / (r (r - 1))], isTrueQ},
{"Identity 3", HoldForm[Sin[γ] Sin[δ] / (Sin[α] Sin[β])],
  HoldForm[s (s - M) / (M (s - 1))], isTrueQ},
{"Identity 4", HoldForm[Sin[σ - γ] Sin[σ - δ] / (Sin[σ - α] Sin[σ - β])],
  HoldForm[(s - M) / (s (s - 1))], isTrueQ},
{"Identity 5", HoldForm[Sin[α] Sin[γ] / (Sin[β] Sin[δ])],
  HoldForm[f (f - M) / (M (f - 1))], isTrueQ},
{"Identity 6", HoldForm[Sin[σ - α] Sin[σ - γ] / (Sin[σ - β] Sin[σ - δ])],
  HoldForm[(f - M) / (f (f - 1))], isTrueQ}};
(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row[{desc_, lhs_, rhs_, test_}] :=
  {desc, Item[TraditionalForm[lhs], Alignment → Center],
  Item[TraditionalForm[rhs], Alignment → Center],
  If[test[ReleaseHold@lhs, ReleaseHold@rhs],
    Style["Verified", Darker@Green, Bold], Style["FAILED", Red, Bold]]};
Grid[Join[{{Style["Description", Bold], Style["Left Hand Side (LHS)", Bold],
  Style["Right Hand Side (RHS)", Bold], Style["Equality Status", Bold]}},
  row/@identities], Frame → All, Background → {None, {1 → LightGray}},
  Spacings → {2, 2.5}, Alignment → {{Left, Center, Center, Center}},
  ItemStyle → Directive[FontFamily → "Helvetica", 12]]
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Out[155]=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Identity 1	$\frac{\sin(\alpha) \sin(\delta)}{\sin(\beta) \sin(\gamma)}$	$\frac{r(r-M)}{M(r-1)}$	Verified
Identity 2	$\frac{\sin(\sigma-\alpha) \sin(\sigma-\delta)}{\sin(\sigma-\beta) \sin(\sigma-\gamma)}$	$\frac{r-M}{r(r-1)}$	Verified
Identity 3	$\frac{\sin(\gamma) \sin(\delta)}{\sin(\alpha) \sin(\beta)}$	$\frac{s(s-M)}{M(s-1)}$	Verified
Identity 4	$\frac{\sin(\sigma-\gamma) \sin(\sigma-\delta)}{\sin(\sigma-\alpha) \sin(\sigma-\beta)}$	$\frac{s-M}{s(s-1)}$	Verified
Identity 5	$\frac{\sin(\alpha) \sin(\gamma)}{\sin(\beta) \sin(\delta)}$	$\frac{f(f-M)}{M(f-1)}$	Verified
Identity 6	$\frac{\sin(\sigma-\alpha) \sin(\sigma-\gamma)}{\sin(\sigma-\beta) \sin(\sigma-\delta)}$	$\frac{f-M}{f(f-1)}$	Verified

Section 3

Results of Section 3 are used in the proof of Remark 12 (see paper)

In[136]:=

```
(* =====*)
(*PART 1:INITIALIZATION& DOMAIN ASSUMPTIONS*)
(* =====*)
ClearAll[α, β, γ, δ, σ, a, b, c, d, r, s, M, assumptions, isTrueQ, row, identities];
σ = (α + β + γ + δ) / 2;
a = Sin[α] / Sin[σ - α]; b = Sin[β] / Sin[σ - β];
c = Sin[γ] / Sin[σ - γ]; d = Sin[δ] / Sin[σ - δ];
r = a d; s = c d; M = a b c d;
(*Domain assumptions:the flat angles lie in (0,Pi)*)
assumptions = {0 < α < Pi, 0 < β < Pi, 0 < γ < Pi, 0 < δ < Pi};
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions]];
(* =====*)
(*PART 2:THE TWO CONVERSION IDENTITIES*)
(* =====*)
identities = {"Identity for sin(α)", HoldForm[Sin[α]],
  HoldForm[(r Sin[σ - α] / (s Sin[σ - γ])) Sin[γ]], isTrueQ},
  {"Identity for sin(β)", HoldForm[Sin[β]],
  HoldForm[(M Sin[σ - β] / (r s Sin[σ - δ])) Sin[δ]], isTrueQ}};
(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row[{desc_, lhs_, rhs_, test_}] :=
  {desc, Item[TraditionalForm[lhs], Alignment → Center],
  Item[TraditionalForm[rhs], Alignment → Center],
  If[test[ReleaseHold@lhs, ReleaseHold@rhs],
  Style["Verified", Darker@Green, Bold], Style["FAILED", Red, Bold]]};
Grid[Join[{{Style["Description", Bold], Style["Left Hand Side (LHS)", Bold],
  Style["Right Hand Side (RHS)", Bold], Style["Equality Status", Bold]}},
  row /@ identities], Frame → All, Background → {None, {1 → LightGray}},
  Spacings → {2, 2.5}, Alignment → {{Left, Center, Center, Center}},
  ItemStyle → Directive[FontFamily → "Helvetica", 12]]
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Out[145]=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Identity for $\sin(\alpha)$	$\sin(\alpha)$	$\frac{(r \sin(\sigma - \alpha)) \sin(\gamma)}{s \sin(\sigma - \gamma)}$	Verified
Identity for $\sin(\beta)$	$\sin(\beta)$	$\frac{(M \sin(\sigma - \beta)) \sin(\delta)}{r s \sin(\sigma - \delta)}$	Verified

Section 4

Results of Section 4 are used in the proof of Lemma 13 (see paper)

In[161]:=

```
(* =====*)
(*PART 1:INITIALIZATION*)
(* =====*)
ClearAll[f, l, m, M, isTrueQ, row, identities];
(*The identities below are identities of
rational functions in Subscript[f,l],Subscript[f,m],M;
the subscripted symbols remain inert,so no domain assumptions are needed*)
isTrueQ[lhs_, rhs_] := TrueQ[Simplify[lhs == rhs]];
(* =====*)
(*PART 2:THE FOUR FACTORIZATION IDENTITIES*)
(* =====*)
identities = {"Factorization 1",
  HoldForm[Subscript[f, l] (Subscript[f, l] - M) / (M (Subscript[f, l] - 1)) -
    Subscript[f, m] (Subscript[f, m] - M) / (M (Subscript[f, m] - 1))],
  HoldForm[(Subscript[f, l] - Subscript[f, m])
    (Subscript[f, l] Subscript[f, m] - (Subscript[f, l] + Subscript[f, m]) + M) /
    (M (Subscript[f, l] - 1) (Subscript[f, m] - 1))],
  isTrueQ}, {"Factorization 2", HoldForm[
  (Subscript[f, l] (Subscript[f, l] - M) / (M (Subscript[f, l] - 1))) *
    (Subscript[f, m] (Subscript[f, m] - M) / (M (Subscript[f, m] - 1))) - 1],
  HoldForm[(Subscript[f, l] Subscript[f, m] - M) (Subscript[f, l]
    Subscript[f, m] - M (Subscript[f, l] + Subscript[f, m] - 1)) /
    (M^2 (Subscript[f, l] - 1) (Subscript[f, m] - 1))], isTrueQ},
{"Factorization 3", HoldForm[(Subscript[f, l] - M) /
  (Subscript[f, l] (Subscript[f, l] - 1)) -
  (Subscript[f, m] - M) / (Subscript[f, m] (Subscript[f, m] - 1))],
  HoldForm[(Subscript[f, m] - Subscript[f, l]) (Subscript[f, l]
    Subscript[f, m] - M (Subscript[f, l] + Subscript[f, m] - 1)) /
    (Subscript[f, l] Subscript[f, m] (Subscript[f, l] - 1)
    (Subscript[f, m] - 1))], isTrueQ}, {"Factorization 4",
  HoldForm[((Subscript[f, l] - M) / (Subscript[f, l] (Subscript[f, l] - 1))) *
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((Subscript[f, m] - M) / (Subscript[f, m] (Subscript[f, m] - 1))) - 1],
HoldForm[(Subscript[f, l] Subscript[f, m] - M)
(Subscript[f, l] + Subscript[f, m] - Subscript[f, l] Subscript[f, m] - M) /
(Subscript[f, l] Subscript[f, m] (Subscript[f, l] - 1)
(Subscript[f, m] - 1))], isTrueQ}];
(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row[{desc_, lhs_, rhs_, test_}] :=
{desc, Item[TraditionalForm[lhs], Alignment → Center],
Item[TraditionalForm[rhs], Alignment → Center],
If[test[ReleaseHold@lhs, ReleaseHold@rhs],
Style["Verified", Darker@Green, Bold], Style["FAILED", Red, Bold]]};
Grid[Join[{{Style["Description", Bold], Style["Left Hand Side (LHS)", Bold],
Style["Right Hand Side (RHS)", Bold], Style["Equality Status", Bold]}},
row/@identities], Frame → All, Background → {None, {1 → LightGray}},
Spacings → {2, 2.5}, Alignment → {{Left, Center, Center, Center}},
ItemStyle → Directive[FontFamily → "Helvetica", 12]]

```

Out[165]=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Factorization 1	$\frac{f_l (f_l - M)}{M (f_l - 1)} - \frac{f_m (f_m - M)}{M (f_m - 1)}$	$\frac{(f_l - f_m) (f_l f_m - (f_l + f_m) + M)}{M (f_l - 1) (f_m - 1)}$	Verified
Factorization 2	$\frac{(f_l (f_l - M)) (f_m (f_m - M))}{(M (f_l - 1)) (M (f_m - 1))} - 1$	$\frac{(f_l f_m - M) (f_l f_m - M (f_l + f_m - 1))}{M^2 (f_l - 1) (f_m - 1)}$	Verified
Factorization 3	$\frac{f_l - M}{f_l (f_l - 1)} - \frac{f_m - M}{f_m (f_m - 1)}$	$\frac{(f_m - f_l) (f_l f_m - M (f_l + f_m - 1))}{f_l f_m (f_l - 1) (f_m - 1)}$	Verified
Factorization 4	$\frac{(f_l - M) (f_m - M)}{(f_l (f_l - 1)) (f_m (f_m - 1))} - 1$	$\frac{(f_l f_m - M) (f_l + f_m - f_l f_m - M)}{f_l f_m (f_l - 1) (f_m - 1)}$	Verified